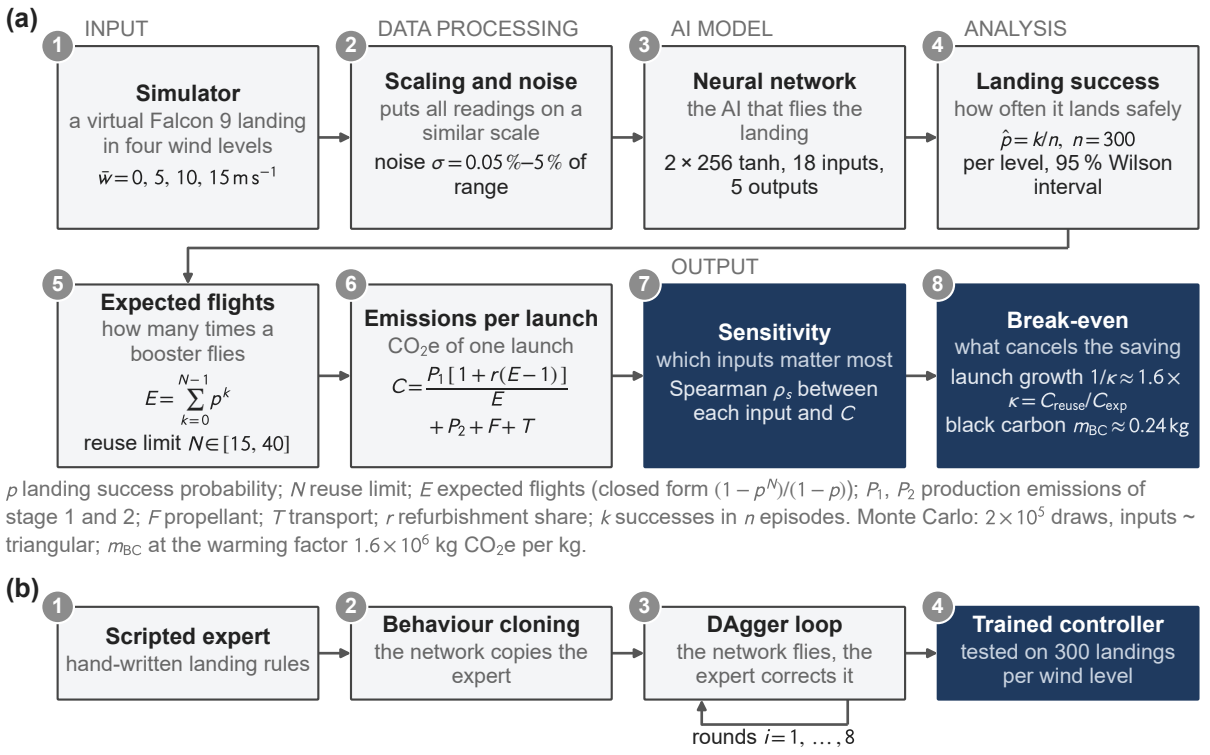


p landing success probability; N reuse limit; E expected flights (closed form $(1 - p^N)/(1 - p)$); P_1, P_2 production emissions of stage 1 and 2; F propellant; T transport; r refurbishment share; k successes in n episodes. Monte Carlo: 2×10^5 draws, inputs ~ triangular; m_{BC} at the warming factor 1.6×10^6 kg CO_2e per kg.